

Design proposal of a Silicon Photonics Mach Zehnder filter:

practical implementation and some theoretical aspects

Ziyad BENKHADAJ
edX username: z-benkhadaaj

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"This report is my own original work and contains no plagiarism."

1 Introduction

Photonic Integrated Circuits are a rapidly growing high technology field with multiple applications and benefits for society. Unintuitively enough, Silicon, despite all its drawbacks and weak physical properties when it comes to optics, is a rather good candidate when it comes to integrated photonics [1]. Particularly, these circuits promise significant progress in optical telecommunication, data communication specifically in datacenters, medical applications as well as more exciting fields like quantum technologies... In this proposal, I will present a basic yet very important building block of photonic circuits: Mach Zehnder Interferometer (MZI). Whether it is for optical filtering, light modulation, variable attenuation or even more complicated optical functions, a MZI is an efficient and rather easy way of implementing those functions in comparison with other solutions.

As shown in the figure 1, a MZ is a simple structure with two arms that each carry a part of the incident light. It is used in this case for filtering light with heating elements to tune the working wavelength.

2 Theory

A MZ consists of a beam splitter (Y coupler or adiabatic coupler) that divides incoming light into two separate paths of equal length or not, respectively balanced or imbalanced. These

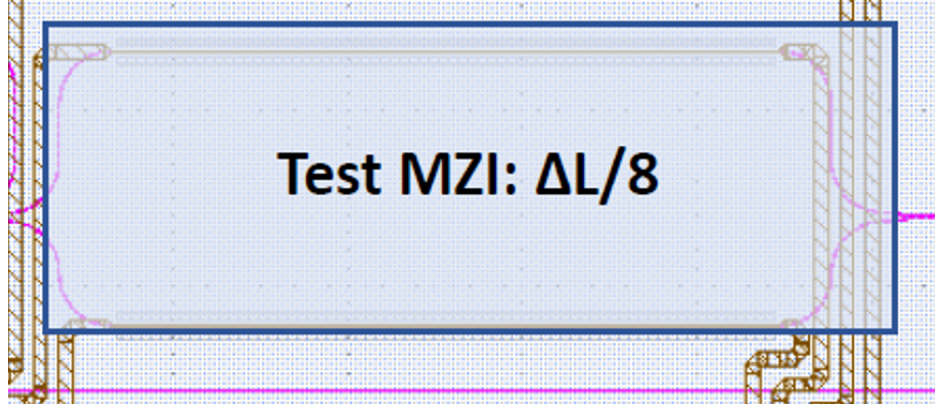


Figure 1: GDS View of a test MZ with two branches used for filtering. I used this device for previous work. In this figure, we can see in pink the waveguides (here done in Silicon Nitride) and in dashed yellow metallic structures used to heat and tune its behavior. Typical scale of the image is hundreds of microns.

paths recombine again at a beam combiner, creating an interference pattern depending on the wavelength and the length difference. To alter the interference pattern, the system can be adjusted so that the paths are different lengths, causing a phase shift between the beams. Additional adjustments can be made using thermo-optic phase shifters, which modify the waveguide's refractive index (of the core and cladding and thus the effective index n_{eff}) by changing the temperature. Light also experiences attenuation throughout the propagation which results in losses in the response function.

Given an input light with intensity I_i and a complex electric field E_i , the beam is split equally (theoretically but we can also account for an imbalance) between both arms. Each arm carries half the initial intensity:

$$I_1 = I_2 = \frac{I_i}{2}$$

and has an electric field :

$$E_1 = -jE_2 = \frac{E_i}{\sqrt{2}}$$

For a lossless system where light propagates through the interferometer arms, the electric fields at the beam combiner inputs are:

$$E_{1p} = E_1 e^{-j\beta L_1}$$

after propagation in L_1 (length of the arm 1), and:

$$E_{2p} = E_2 e^{-j\beta L_2}$$

after propagation in L_2 (length of the arm 2), β being the propagation constant of the lightwave in the waveguide and using the optician convention.

The electric field and consequently the intensity at the output of the MZ is:

$$I_{output} = (jE_{1p} + E_{2p}) \cdot (jE_{1p} + E_{2p})^* = I_i \cos^2\left(\frac{\beta(L_1 - L_2)}{2}\right)$$

where the star sign $*$ is for complex conjugation.

The latter function has a periodic behavior with respect to the wavelength λ . Putting $\Delta L = L_2 - L_1$ and $\beta = \frac{2\pi n_{eff}(\lambda)}{\lambda}$ and knowing that cosine squared is π periodic, we have that the wavelengths reproducing the same output intensity are defined as follows for every p integer:

$$\frac{n_{eff}(\lambda_p)}{\lambda_p} = \frac{p}{\Delta L}$$

and considering two closest wavelengths for consecutive orders, we can approximate the difference of wavelength (FSR) using a first order derivation (of the function: $f(\lambda) = \frac{n_{eff}(\lambda)}{\lambda}$):

$$FSR = \frac{\lambda^2}{n_g \Delta L}$$

and $n_g = n_{eff}(\lambda) - \lambda n'_{eff}(\lambda)$.

3 Modelling and simulation

For designing the waveguides, we use a core of Silicon of approximately 220 nm of thickness and 500 nm of width for the different reasons discussed in the course (to avoid multimode behavior, industry standard...). This core is burried in a cladding of oxide a little bit less refringent as usually used in photonic circuits. Unless otherwise specified, I worked with TE modes only.

We use a simulation software from ANSYS, namely Lumerical, in order to perform simulations on the waveguides that were designed. One of the goals of these simulations are to determine the supported modes (TE, TM in particular), in the structure, especially the fundamental modes, and find the propagation constant and the associated effective and group index as a function of the wavelength. I show some important figures in 2, 3, 4. It is interesting feature to see for example is that while it is known that the effective index is generally (for many materials) a decreasing function of the wavelength, the group index is, on the other hand, an increasing function of the wavelength.

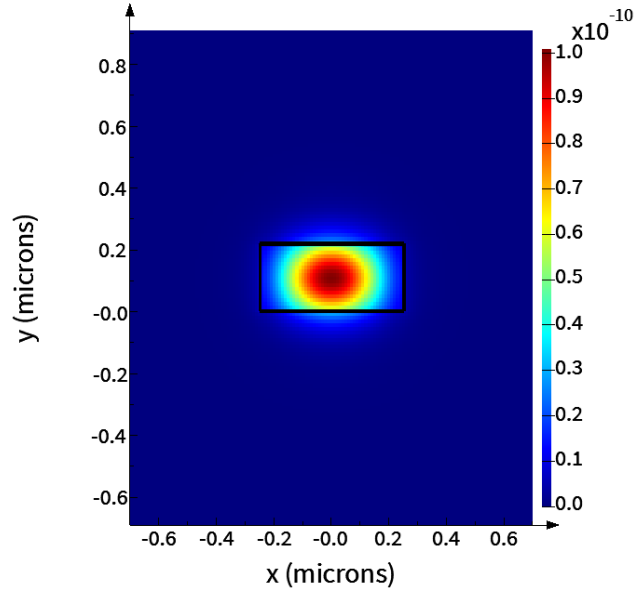


Figure 2: Energy density of the TE mode. Scale is linear and in arbitrary energy units.

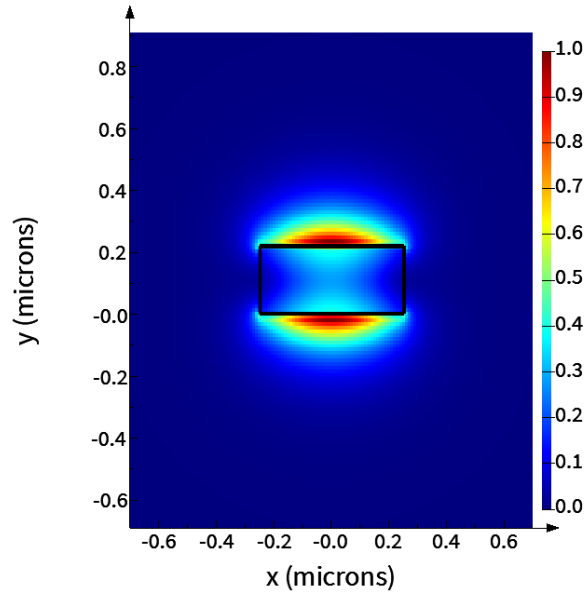


Figure 3: Energy density of the TM mode. Scale is linear and in arbitrary energy units.

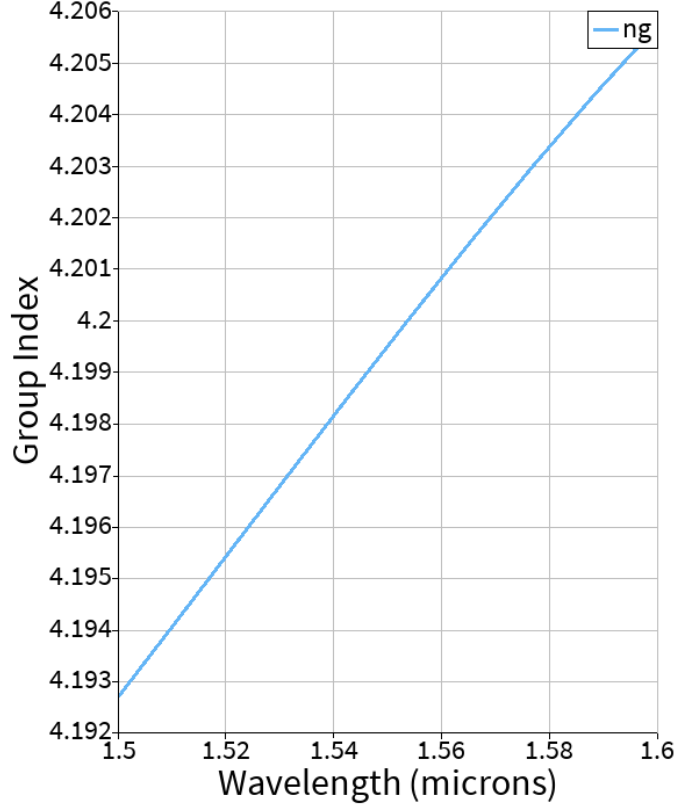


Figure 4: Group index profile of the fundamental TE mode as a function of the wavelength

We can determine from this work the effective index's dependence as a function of the wavelength, in the form of a Taylor expansion near a central wavelength of $\lambda_0 = 1.55\mu m$. For the fundamental TE mode, the coefficients found after interpolation (curve fitting) enable to write the following expression:

$$n_{eff}(\lambda) = 2.44543 - 1.13161(\lambda - \lambda_0) - 0.04260(\lambda - \lambda_0)^2$$

In order to show the effect of parameter variation, it is also interesting to see the behavior of the MZ spectrally for different path differences, as shown in 5

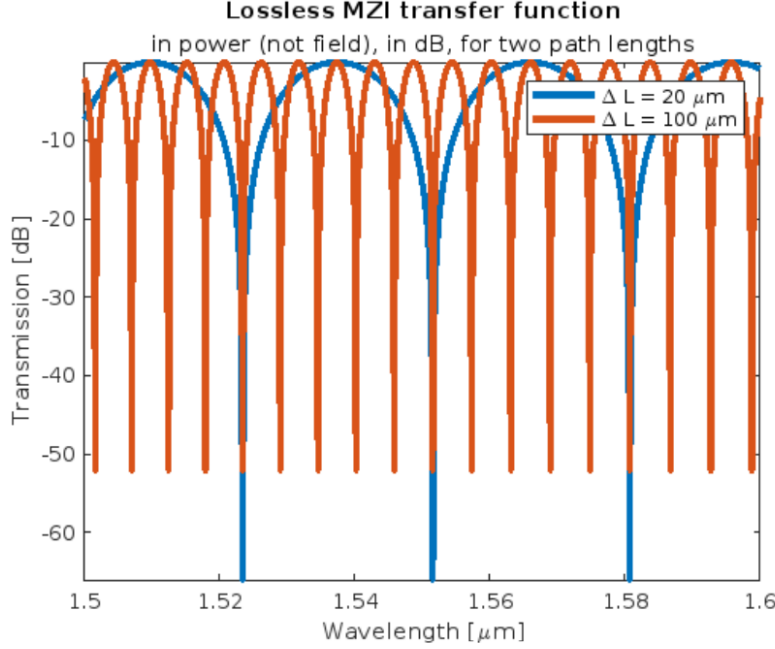


Figure 5: Spectral behavior of an imbalanced MZ for two different path length differences.

4 Layout design

In this section, I will present the design I submitted for fabrication. The design was done using KLayout and the SiEPIC PDK thanks to the instructions of the course. The idea to create a cascade of MZs with different arms lengths imbalances in order to create a filtering function.

4.1 Theoretical aspects

Consider an imbalanced MZ with arms lengths imbalance: ΔL_1 . The ideal spectral response (in power) considering no insertion loss and no spectral dependence of the entry and exit of the circuit is:

$$f_1(\lambda) = \cos^2\left(\frac{\beta(\lambda)\Delta L_1}{2}\right)$$

with periodicity: $FSR_1 = \frac{\lambda_0^2}{n_g \Delta L_1}$

Now, we consider a second and third MZs each with lengths imbalances: $\Delta L_2 = 2\Delta L_1$ and $\Delta L_3 = 2\Delta L_2 = 4\Delta L_1$. The individual responses of the MZs follow the ideal responses: $f_i(\lambda) = \cos^2\left(\frac{\beta(\lambda)\Delta L_i}{2}\right)$ with $i = 1, 2, 3$ as shown in the figure 6.

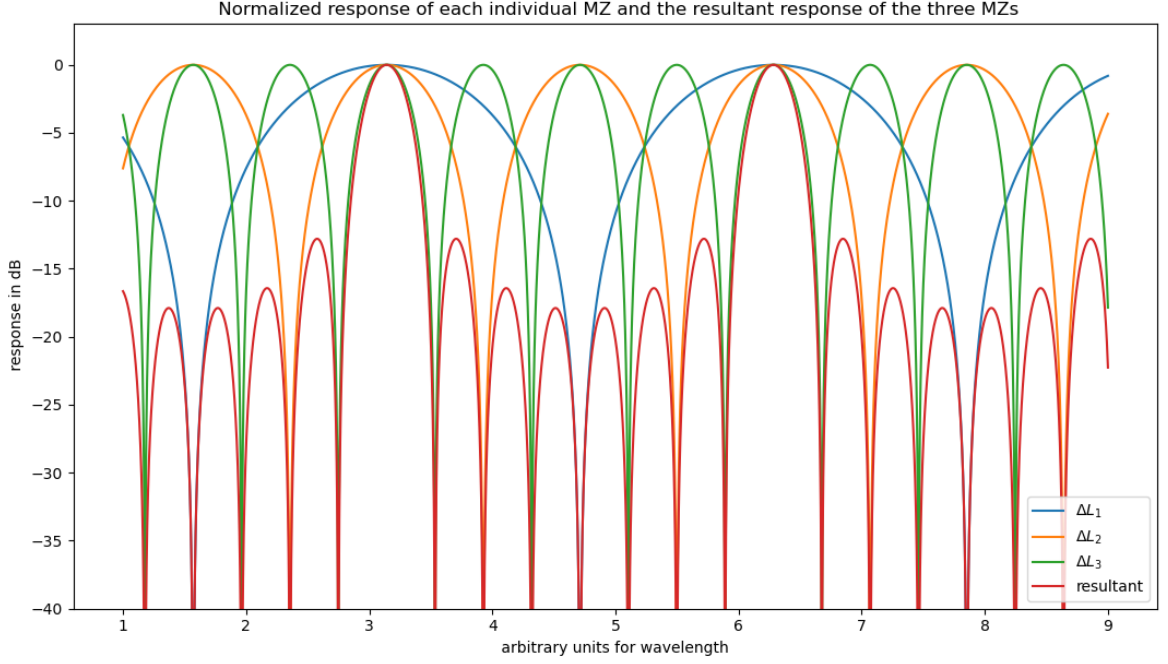


Figure 6: Spectral behavior of a cascade of imbalanced MZs.

Of course and as we will see later, the spectral responses of the 3 MZs will suffer from mismatch due to variations of the fabrication. This variability can be modeled as a wavelength shift in the spectral domain, by using the following responses:

$$f_i(\lambda) = \cos^2\left(\frac{\beta(\lambda - \lambda_i)\Delta L_i}{2}\right)$$

where λ_i is the shift accounted for spectrally, with respect to the ideal case.

For a certain triplet of λ_i , we can simulate the response obtained, as shown in the figure 7.

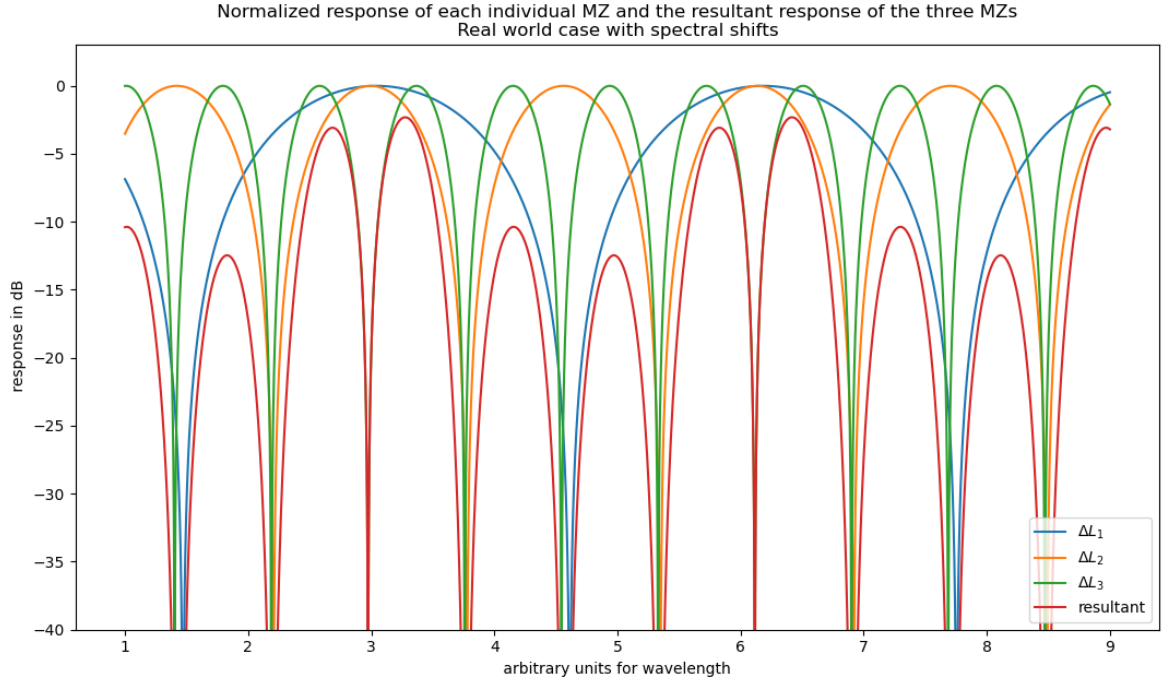


Figure 7: Spectral behavior of a cascade of imbalanced MZs accounting for spectral shifts.

4.2 Design and layout

This part consists of the design and layout of the desired circuit using the useful softwares. In my case, I used KLayout for design and layout, with the help of the SiEPIC PDK. The simulation part was done with MODE and Interconnect. The layout submitted is in figure 8. We see that I have reproduced 8 similar structures of the circuit I presented earlier. Please note that the entry of each circuit is at the second (from top to bottom) grating coupler. On the other hand, the output grating couplers are labeled Channel or Detector (as we will see in the following) 1, 2, 3 from bottom to top.

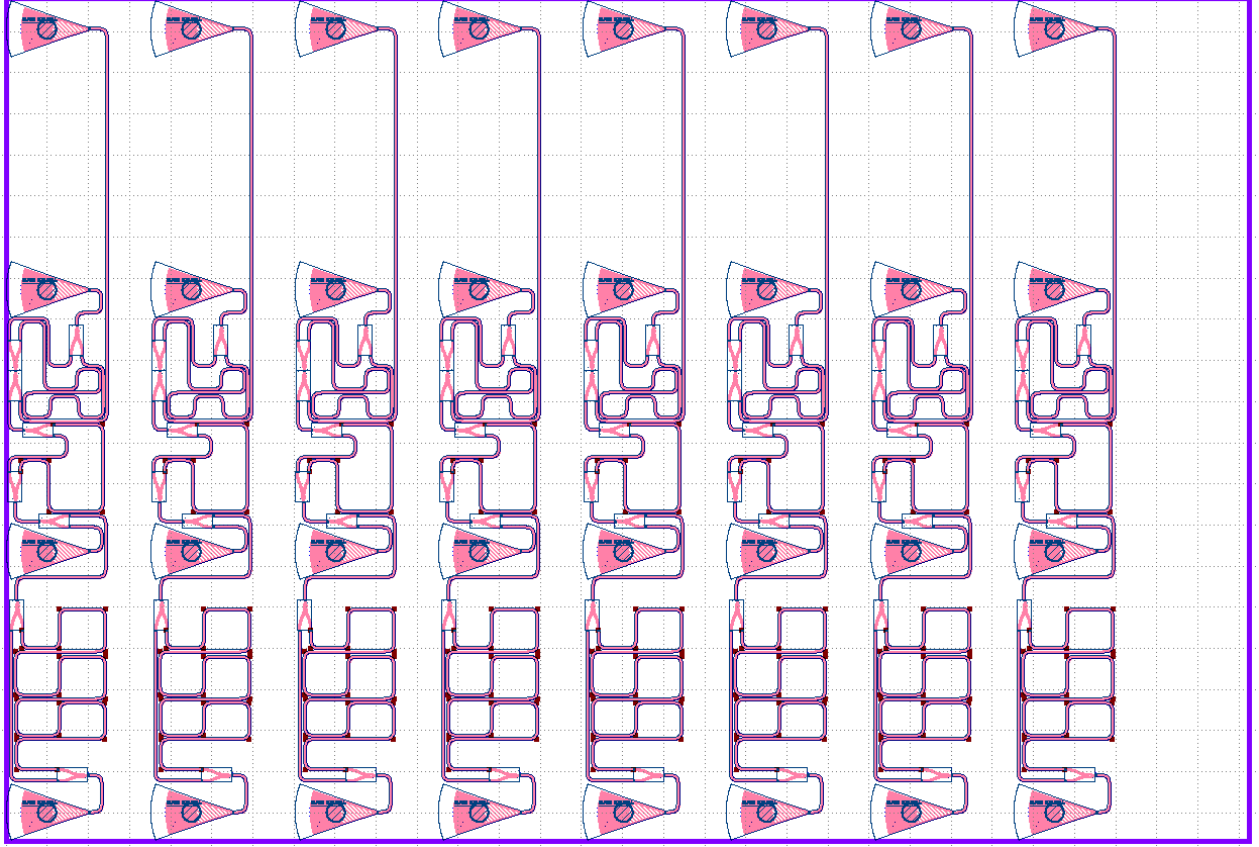


Figure 8: Layout of the circuit in KLayout. Please note that the grating couplers from top to bottom correspond to the following: Detector 3, Entry laser, Detector 2, Detector 1

5 Fabrication

In this section, we will briefly understand the fundamental steps of fabrication of photonic integrated circuits. In particular, we will see how this is done in the Applied Nanotools, Inc. facilities in Edmonton, CANADA. Alternatively, the Washington Nanofabrication Facility (WNF) has a similar fabrication process optimized for different geometries (e.g. thicker wafers) and solvents.

5.1 Some key aspects

The fabrication of photonic circuits is to some extent similar to electronic circuits. It consists of a lithography process to write specific patterns on a wafer, silicon in our case. One key aspect in this operation, is the minimum pattern size that can be written on the wafer surface. There are many sources of limitation like the writing layer thickness, the resist quality, the technology used...etc. Regarding this last aspect, there are two main types of lithography: optical lithography (or photolithography) and electron beam lithography (EBL). Some differences will be explained in the following.

5.2 EBL vs Photolithography

The difference between the two technologies relies essentially in the nature of physical entities used to perform the patterning of the wafer layers. In one case it is photons, in the other, it is electrons. The following table presents some useful differences between the two technologies:

EBL	Photolithography
No photomask required	Requires photomask
Serial process	Parallel process
Highly flexible (many variations possible on the same wafer)	Rigid (same pattern across wafer)
Strict environmental constraints (e.g., pressure, vacuum)	Fewer environmental constraints
Fast and easy for complex/small patterns	Slower and more difficult

6 Experimental data and analysis

In this section, I will expose the results of the measurements carried out by my instructors and discuss the relevance of these measurements.

Please note that the filtering function that was proposed in the design was not clearly visible in the results due to considerations that we will see together here.

Following the nomenclature that we saw before for the ordering of the detectors, we can see in the figure 9 the results for one of the 8 circuits presented in 8. Those results were extracted using Python from the CSV files received, and then analysed.

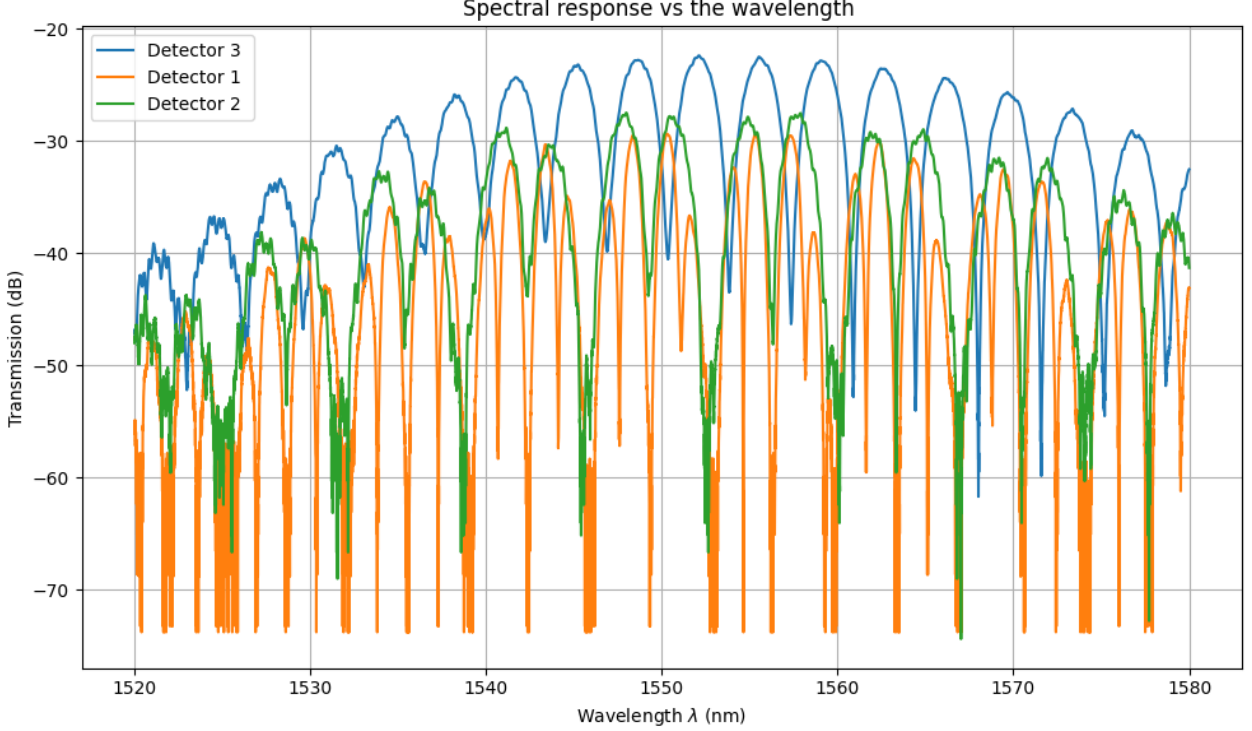


Figure 9: Spectral response of the three detectors for one of the 8 circuits of the layout. Note that each detector issues a different combination of the three MZs.

The Detector 1 corresponds to the cascade of the three MZs. The spectral response is far from the expected result (c.f 6). This is due the spectral shifts of each of the three MZs as discussed earlier. We can also see that the Detector 2 also suffers from the same problem, being the product of two MZs.

For the analysis I conducted, I used Python scripts to extract and analyze the data from the CSV files of each of the 8 circuits that were measured. Some useful parts of the script are shown in 10.

In this part, I focused on the Detector 3 consisting of only one MZ in the layout since the filtering function is not realized probably due to mismatch between the spectral response of the three MZs as explained before. Also, I focus on the wavelength range between $1520nm$ and $1580nm$ symmetrically around $1550nm$.

The analysis is performed through the following steps:

i) Conditioning of the wavelength range and dynamic range ii) Baseline correction using a polynomial fit to account for the grating coupler response iii) Extraction of useful data: FSR, Extinction ratio...

```

main.py  traitement_donnees.py x
1  import matplotlib.pyplot as plt
2  import numpy as np
3  import scipy as sp
4
5  filename = "20-Jun-2025 06.16.21.csv"
6  with open(filename, 'r', encoding='utf-8') as f:
7      lines = f.readlines()
8
9  nom_device = lines[5]
10 resolution = float(lines[14].split('\t')[2][:5])
11 lam_debut = lines[15]
12 lam_fin = lines[16]
13 wavelengths = lines[23].split(',')[1:]
14 channel1 = lines[24].split(',')[1:]
15 channel2 = lines[25].split(',')[1:]
16 channel3 = lines[26].split(',')[1:]
17 channel4 = lines[27].split(',')[1:]
18
19 N = int(0.25 * len(wavelengths)) # we ignore the 25% first values for 2 reasons:
20                                     # so that the wavelength range is centered around 1550 nm
21                                     # avoid noisy signal

```

```

main.py  traitement_donnees.py x
30  ##### Baseline correction
31
32  degre = 5 # best value in my case
33  lamda_um = wavelengths/1e3 # to convert to microns
34  coeff = np.polyfit(lamda_um, channel3, degre, rcond=10)
35  polynome = np.poly1d(coeff)
36  fit = polynome(lamda_um)
37  val = channel3 - fit
38  val = val - np.max(val)
39
40  peaksmx, _ = sp.signal.find_peaks(val, distance=600)
41  peaksmin, _ = sp.signal.find_peaks(-val, distance=600)
42  FSRmax = np.diff(peaksmx)*resolution # based on maxima
43  FSRmin = np.diff(peaksmin)*resolution # based on minima
44  FSR = np.concatenate((FSRmin, FSRmax)) # in order to take into account both methods in the computation
45  #print(" FSR : ", FSR )

```

Figure 10: some useful functions for data analysis

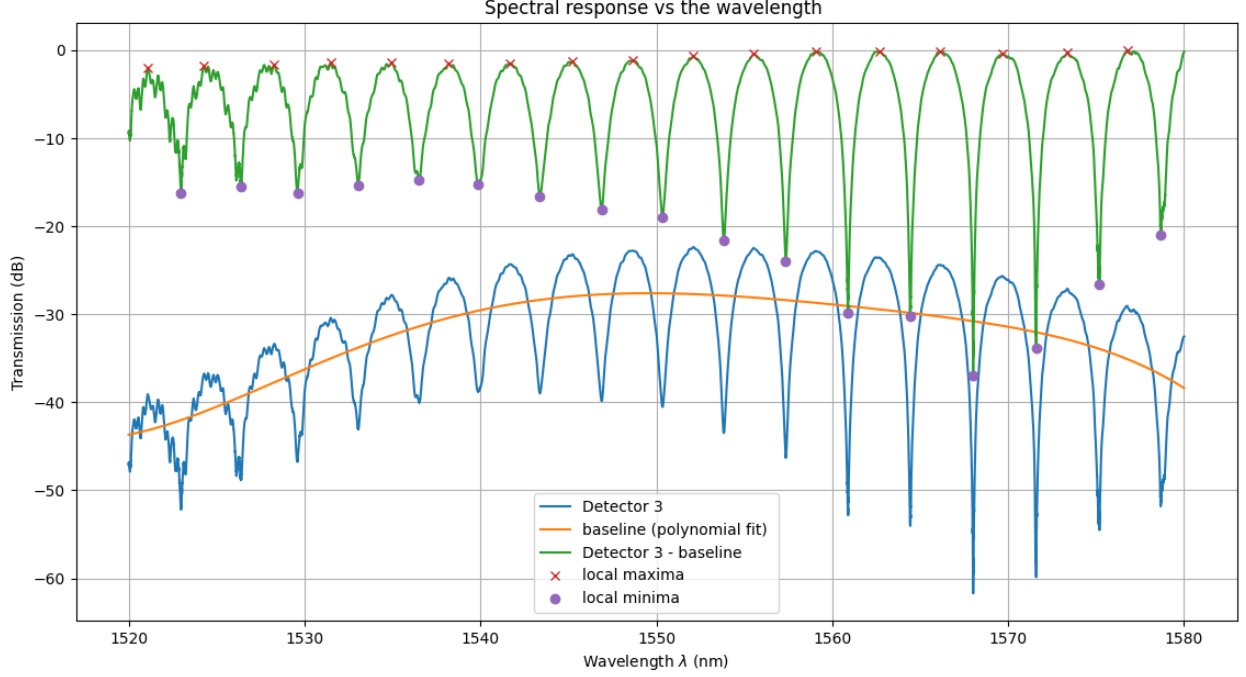


Figure 11: The spectral response of the Detector 3 (upper grating coupler in the layout) which has been fitted and local maxima and minima isolated to extract FSR and group index.

In the figure 11, an example is given showing the spectral response of one of the circuits. The Detector 3 in particular is shown in blue. The effect of the grating couplers is accounted for and removed by fitting a 5th order polynomial (that's what worked the best for me). The resulting response is then normalised such that its maximum lies at 0 dB. The local maxima and minima can then be isolated using different methods (I used `scipy's signal.find_peaks()`). This enables the extraction of many pieces of information. and compare with the values obtained (for the group index for example) with other simulation methods such as the corner analysis, which is presented here.

7 Comparison: Simulation vs Experimental

In this section, I compared the results obtained through simulation using corner analysis to the experimental results obtained with the previous analysis part. I also used MonteCarlo analysis and the variability was very similar to the corner analysis. In my case, I considered MODE to perform simulation while varying the width and depth of the waveguide. The scheme in figure 12 summarizes the performed work. In order to be concise and clear, are shown the 4 corners and center that give enough variability to find the maximum and minimum values for the desired parameters. I find the following at 1550 nm:

- **A:** $n_g = 4.26$ and $n_{eff} = 2.40$
- **B:** $n_g = 4.18$ and $n_{eff} = 2.47$

- **C**: $n_g = 4.17$ and $n_{eff} = 2.44$
- **D**: $n_g = 4.24$ and $n_{eff} = 2.37$
- **E**: $n_g = 4.21$ and $n_{eff} = 2.42$

On the other hand, the analysis performed earlier gives a mean value: $n_g = 4.21$ with a standard deviation: $\sigma \approx 0.052$ taking into account the $\sqrt{n}$ and Student coefficient.

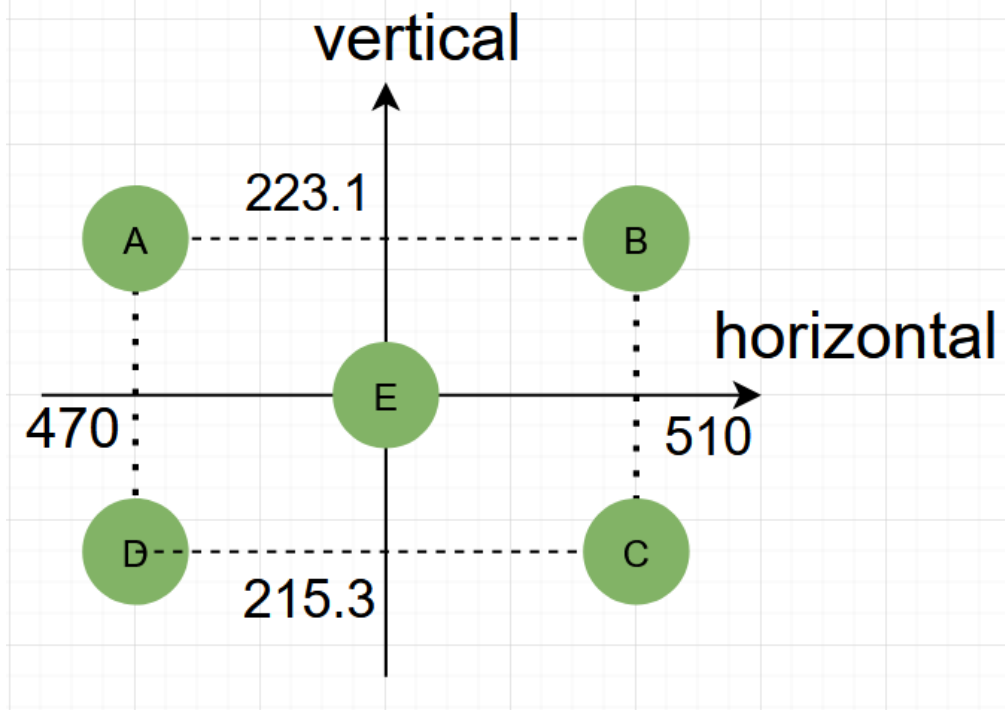


Figure 12: The units are in nm. The five locations in green are the corners in which the simulation in MODE's FDE was performed.

8 Conclusion and acknowledgments

First, I sincerely thank my instructors in this course and the Applied Nanotools facilities that enabled all of this as well as the Canadian photonics consortium SiEPIC. I thank Omid Esmaeeli for performing the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

This introductory course on the design, modeling, simulation, and fabrication of photonic integrated circuits gave me a modestly good understanding of the entire development workflow and the main challenges involved in producing a circuit.

I am a little bit disappointed by the behaviour of the cascaded MZs which is different from the expected one. Nevertheless, the analysis conducted enabled me to confirm the theoretical and simulation predictions, it also trained me beneficially in data analysis.

I worked with approximately 220×500 nm waveguide and implemented TE MZ circuits repeated multiple times to assess repeatability. The experimental results suggested that repeatability was not the primary issue, even though MonteCarlo and corner analysis simulations indicated significant variability.

When comparing simulated and experimental data, it became clear that certain parameters like wavelength-dependent loss, extinction ratio, and wavelength shifts are difficult to predict precisely. However, the simulation results for FSR and group index showed strong agreement with experimental observations for TE modes.

References

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